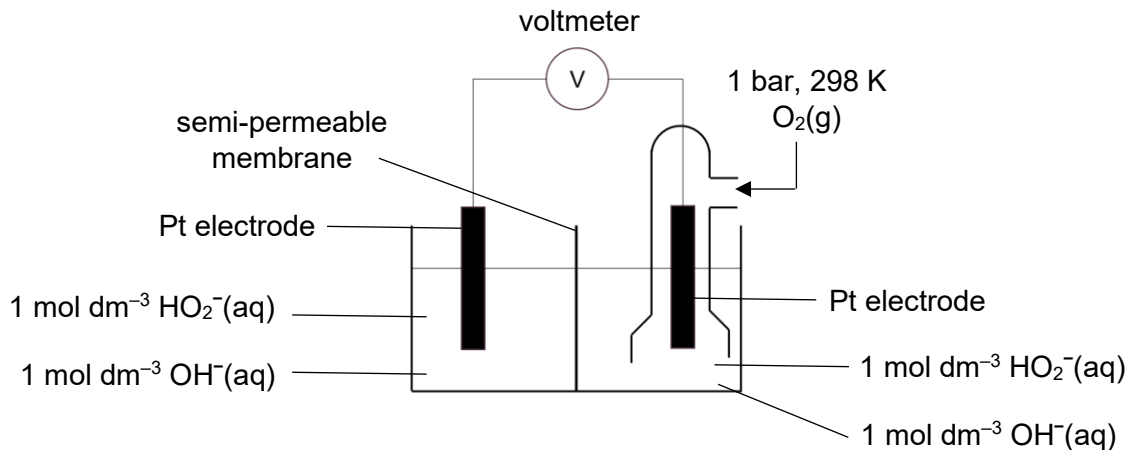


14 Use of the Data Booklet is relevant to this question

Hydrogen peroxide deprotonates in alkaline medium to form HO_2^- . The diagram below shows an electrochemical cell set up to measure the cell potential for the decomposition of HO_2^- to form OH^- and O_2 gas.



Which statements are correct?

- 1 The voltmeter gives a reading of -0.96 V .
 - 2 The reaction at the cathode may be given as $\text{HO}_2^- + \text{H}_2\text{O} + 2\text{e}^- \rightarrow 3\text{OH}^-$.
 - 3 The semi-permeable membrane behaves like a salt bridge in the cell.
- A** 1 only
- B** 1 and 3 only
- C** 2 and 3 only
- D** 1, 2 and 3 only